

PROJECT REPORT CONTENT

Statewide Safety, Protection & Wellbeing Programme for KGBVs & NSCBAVs, Madhya Pradesh

Power BI-Based Programme Analytics & Decision Support Solution

Project Type: Data Analytics / Business Intelligence / Dashboarding

Primary Platform: Microsoft Power BI

Core Technologies: Power BI, Power Query, DAX, Excel

Assessment Model: Pre-Assessment → Programme Modules → Post-Assessment → Learning Gain

1. Executive Summary

The **Statewide Safety, Protection & Wellbeing Programme for KGBVs & NSCBAVs, Madhya Pradesh** is an analytics and decision-support solution designed to monitor programme participation, assessment performance, completion and learning outcomes across participating institutions, districts and programme levels.

The solution transforms assessment-response data into a structured analytical model and an interactive Power BI dashboard. The reporting framework covers both **students and personnel**, enabling analysis at state, district and institution levels.

The analytical approach uses a **star-schema data model** containing shared dimensions for dates, districts, institutions, modules and questions, supported by separate student and personnel response facts.

The dashboard has evolved into a six-page hierarchy:

1. Home / Dashboard Overview
2. State Executive
3. State Student
4. State Personnel
5. Institution Student
6. Institution Personnel

The final dashboard design emphasizes executive visibility, geographic analysis, participation monitoring, Pre/Post assessment comparison and learning-gain analysis.

2. Project Overview

2.1 Background

The programme focuses on safety, protection and wellbeing learning for students in **KGBVs and NSCBVs**.

The analytical requirement is to move beyond simply collecting assessment responses and provide a centralized mechanism for understanding:

- Programme participation
- Assessment completion
- Pre-assessment performance
- Post-assessment performance
- Learning gain
- Module-level performance
- District-level performance
- Institution-level performance
- Student and personnel outcomes

2.2 Problem Statement

Assessment information can become difficult to interpret when responses are maintained across multiple datasets and analysed manually.

The project addresses this by establishing a centralized BI layer capable of connecting assessment responses with:

- Students
- Personnel
- Institutions
- Districts
- Modules
- Questions
- Assessment dates

The resulting dashboard provides interactive monitoring instead of relying solely on raw assessment exports.

2.3 Project Objective

The primary objective is to develop a Power BI-based analytics solution that:

- Consolidates assessment data
- Structures data into an analytical model
- Tracks programme participation
- Compares Pre and Post assessment performance

- Calculates learning gain
- Provides state, district and institution-level analysis
- Separates student and personnel analytics
- Supports interactive filtering and navigation
- Provides executive-level programme visibility

2.4 Scope

The implemented analytical scope includes:

Student analytics

- Student participation
- Student responses
- Module performance
- Pre/Post assessment performance
- Learning gain
- District participation
- Institution-level student performance

Personnel analytics

- Personnel participation
- Personnel responses
- Personnel module performance
- Pre/Post assessment performance
- Learning gain
- District and institution analysis

Geographical levels

- State
 - District
 - Institution
-

3. Requirements

3.1 Business Requirements

The dashboard should provide centralized visibility into:

- Programme reach
- Participating institutions
- Active districts
- Student participation

- Personnel participation
- Completion
- Assessment performance
- Learning outcomes

3.2 Functional Requirements

The solution should support:

- Interactive dashboard navigation
- Institution selection
- District filtering
- Module filtering
- Stage filtering
- Class filtering where applicable
- Year/date filtering
- Student and personnel analysis
- Pre/Post comparison
- Learning-gain analysis

3.3 Analytical Requirements

The solution should enable:

- Response-level analysis
- Student-level improvement analysis
- Personnel-level improvement analysis
- Module-level comparison
- District-level comparison
- Institution-level analysis
- Learning outcome analysis

3.4 Reporting Requirements

The dashboard should provide separate reporting views for:

Level	View
State	Executive, Student, Personnel
Institution	Student, Personnel
Home	Programme overview and navigation

4. Data Sources & Datasets

The project used assessment datasets structured around student and personnel responses.

4.1 Student Assessment Data

The student side contains assessment data for:

- Module 1 — **My Safety, My Rights**
- Module 2 — **Safe & Unsafe Behaviour**

Each module contains Pre and Post assessment responses.

4.2 Personnel Assessment Data

Personnel assessment data contains:

- E1 — **Understanding POCSO & Child Safeguarding**
- E2 — **POSH & Safe Workplace Practices**

Each personnel module contains Pre and Post assessment responses.

4.3 Key Analytical Fields

Important fields include:

- Student_ID
- Employee_ID
- Institution_ID
- School_Code
- District_ID
- Module_ID
- Question_ID
- Response
- Score
- Stage
- Assessment_Date
- Department
- Hostel_Type
- Class
- Age_Group

4.4 Data Grain

The response fact is maintained at approximately:

Participant × Module × Stage × Question

This enables question-level assessment analysis while allowing aggregation to module, institution, district and state levels.

5. Data Preparation & Cleaning

The data preparation process was implemented through Power Query and structured transformation.

Overall workflow

```
graph TD; A[Raw Assessment Data] --> B[Staging Queries]; B --> C[Cleaning & Standardization]; C --> D[Append / Merge]; D --> E[Dimension Preparation]; E --> F[Fact Table Preparation]; F --> G[Relationship Validation]; G --> H[Power BI Model];
```

Raw Assessment Data

↓

Staging Queries

↓

Cleaning & Standardization

↓

Append / Merge

↓

Dimension Preparation

↓

Fact Table Preparation

↓

Relationship Validation

↓

Power BI Model

Key transformations

Append

Pre/Post and module-specific response datasets were combined into the response fact structure.

Merge

Reference information was brought into dimensions through controlled merges, including institution and district information.

Standardization

Fields such as:

- Module
- Stage
- Institution
- District

- Question

were standardized to support relationships and reporting.

Duplicate handling

Question identifiers were checked for uniqueness before establishing relationships with the question dimension.

Date handling

Assessment dates were linked to the date dimension. The project specifically avoided inventing dates for real assessment data.

Validation

Relationships and data completeness were checked during model development.

6. Data Model

6.1 Modeling Approach

The solution uses a **star-schema approach**.

Shared dimensions provide descriptive context while student and personnel response facts contain assessment transactions.

6.2 Main Dimensions

Dimension	Purpose
Dim_Date	Assessment date analysis
Dim_District	District-level reporting
Dim_Institution	Institution-level reporting
Dim_Module	Module information
Dim_Question	Question and competency metadata
Dim_Student	Student attributes
Dim_Employee	Personnel attributes

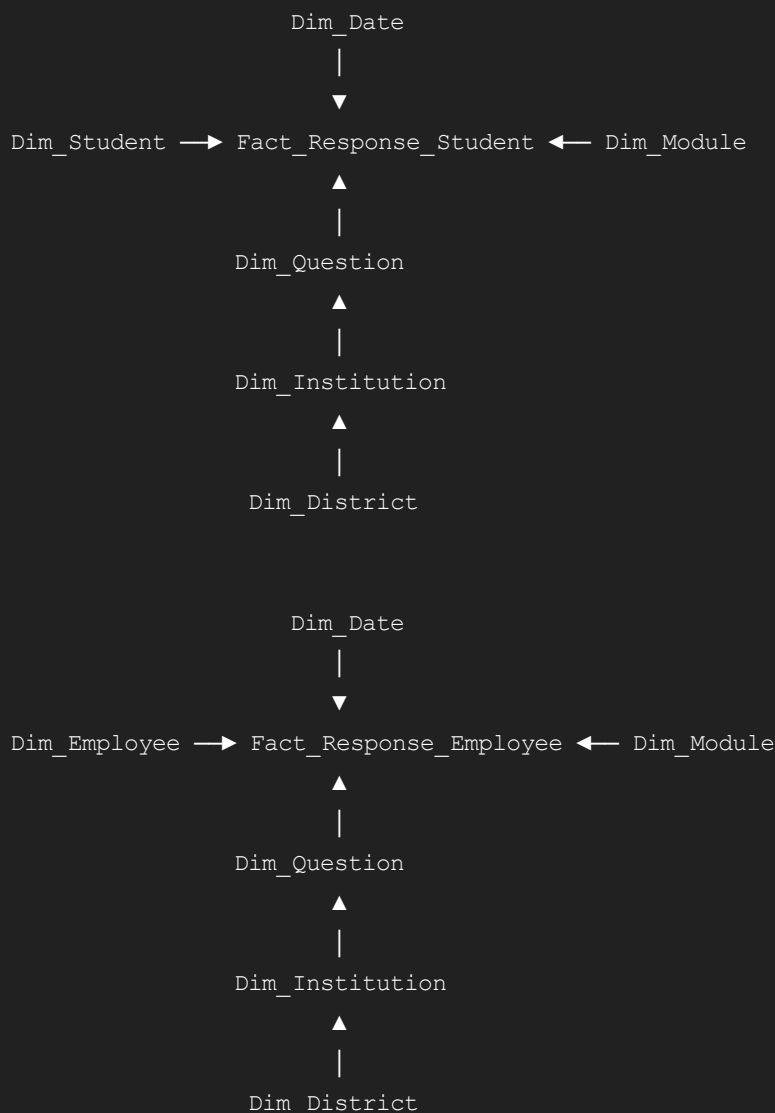
6.3 Main Fact Tables

Fact	Purpose
Fact_Response_Student	Student question-level assessment responses
Fact_Response_Employee	Personnel question-level assessment responses

Additional analytical fact structures were explored for score summaries and feedback, where applicable.

6.4 Relationship Architecture

The intended final architecture follows:



The two response facts remain separate rather than being directly connected to each other.

7. Analytical Logic & DAX

The dashboard uses DAX measures for aggregation, assessment comparison and learning outcomes.

7.1 Students Trained

Conceptually:

Distinct count of participating Student_ID values.

7.2 Personnel Trained

Conceptually:

Distinct count of participating Employee_ID values.

7.3 Response Count

Response count is based on records in the respective response fact table.

7.4 Pre-Assessment Score

The Pre-stage assessment responses are aggregated to calculate the average/percentage performance.

7.5 Post-Assessment Score

The Post-stage assessment responses are aggregated using the same assessment framework.

7.6 Learning Gain

The project uses:

Learning Gain = Post-Assessment Score – Pre-Assessment Score

The metric is interpreted as a **percentage-point difference** when displayed as a percentage.

For example, a movement from 63% to 92% represents a **29 percentage-point improvement**, not 29%.

7.7 Students Showing Improvement

The project also includes participant-level improvement analysis by comparing a student's Pre and Post performance.

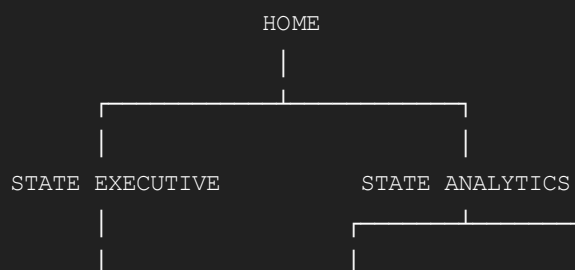
The same analytical concept is applied to personnel.

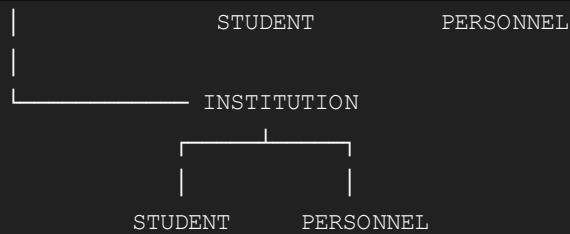
8. KPI Framework

KPI	Definition	Purpose
Institutions Covered	Distinct participating institutions	Programme reach
Districts Active	Districts with programme activity	Geographic coverage
Students Trained	Unique students	Student participation
Personnel Trained	Unique personnel	Personnel participation
Student Completion	Student programme completion	Participation monitoring
Personnel Completion	Personnel completion	Personnel monitoring
Pre Score	Pre-assessment performance	Baseline
Post Score	Post-assessment performance	Final assessment
Learning Gain	Post – Pre	Learning improvement
Response Count	Assessment response records	Assessment activity
Students Showing Improvement	Students with higher Post than Pre score	Individual outcome analysis

9. Dashboard Architecture

The final navigation structure is:





Final Pages

Page 1 — Home / Dashboard Overview

Provides:

- Programme title
- Programme snapshot
- Navigation cards
- Programme flow
- High-level KPIs

Page 2 — State Executive

Provides:

- Madhya Pradesh geographic overview
- Programme reach
- District performance
- Completion
- Learning gain
- Institution-type participation
- Executive insights

Page 3 — State Student

Focuses on:

- Students trained
- Student responses
- Module participation
- Pre/Post performance
- Students showing improvement
- District analysis

Page 4 — State Personnel

Focuses on:

- Personnel trained
- Personnel responses
- Personnel module performance

- Learning gain
- Department analysis
- Key Influencers
- Decomposition Tree

Page 5 — Institution Student

Focuses on:

- Selected institution
- Students trained
- Student completion
- Student response activity
- Student Pre/Post performance
- Learning gain
- Student improvement

Page 6 — Institution Personnel

Focuses on:

- Selected institution
- Personnel trained
- Personnel completion
- Personnel response activity
- Personnel Pre/Post performance
- Learning gain
- Department/module analysis

10. Visualization & UX Design

The final dashboard theme uses a **light blue, white and blue-based professional BI interface**.

The design incorporates:

- KPI cards
- Line charts
- Bar charts
- Column charts
- Donut charts
- Geographic map
- Key Influencers

- Decomposition Tree
- Tables/matrices
- Navigation buttons
- Slicers
- Dynamic institution selection

Design principles

Executive hierarchy

Important information is positioned first:

KPIs → geography → participation → outcomes → detailed analysis

Reduced visual repetition

Student and personnel pages use different combinations of visuals rather than repeating identical chart layouts.

Dynamic institution selection

Institution-level pages use a selected-institution mechanism instead of displaying an unnecessary count of all institutions.

Interactive filtering

Slicers are used to allow users to move from broad programme-level information toward specific analytical views.

11. Interactive Features

Navigation

Power BI buttons are used for page-to-page navigation.

Slicers

Depending on page, filters include:

- District
- Institution
- Hostel Type
- Class
- Module
- Stage
- Year

- Department

Dynamic Institution

The institution pages use a selected institution rather than treating the entire institution population as the primary focus.

Tooltips

Tooltips were considered/used to provide additional context without overcrowding the main dashboard.

Key Influencers

Used on personnel/institution analysis to explore factors associated with the selected analytical outcome.

Decomposition Tree

Used to break down learning gain across dimensions such as:

- Module
 - Department
 - Institution-related attributes
 - Hostel Type
-

12. Analytical Insights

The project demonstrates several levels of analytical capability.

Student level

The dashboard can identify:

- Student participation
- Assessment activity
- Pre/Post performance
- Students showing improvement

Personnel level

The dashboard can analyse:

- Personnel participation
- Module performance
- Pre/Post assessment
- Learning gain
- Department-level patterns

District level

The dashboard enables:

- District participation comparison
- Learning-gain comparison
- Institution coverage analysis

Institution level

The dashboard enables institution-specific:

- Participation
- Completion
- Assessment performance
- Learning outcomes

Important limitation

The demo dataset contains synthetic/sample records. Therefore, dashboard values generated from those demo datasets should not be interpreted as actual programme results.

13. Challenges & Solutions

Challenge	Solution
Multiple assessment datasets	Append response datasets into analytical fact structures
Wide assessment exports	Move toward long-format response modelling
Question dimension duplication	Validate Question_ID uniqueness
Missing/incorrect Stage values	Standardize Pre/Post stage values
Institution/district relationships	Establish separate institution and district dimensions
Date analysis	Create and relate Dim_Date
Employee/personnel integration	Build separate personnel response fact
Completion showing 100%	Treat completion as a calculated analytical result rather than altering source data

Challenge	Solution
Key Influencers initially finding no influencers	Adjusted the Explain-by fields and analytical configuration
Visual repetition	Introduced line charts, map, Key Influencers and Decomposition Tree
Institution card showing “All Institutions”	Align slicer and selected-institution measure with the same institution dimension
Dashboard clutter	Reorganize pages by analytical purpose

14. Validation & Quality Assurance

Validation activities included:

- Checking fact-table row presence
- Checking dimension relationships
- Checking Question_ID uniqueness
- Checking date relationships
- Checking institution-district mapping
- Checking slicer interactions
- Checking Pre/Post measures
- Checking learning-gain calculations
- Checking dynamic institution selection
- Checking dashboard visual behaviour

Where the project documentation does not explicitly establish production-level validation, the result should be considered a development/demo validation rather than a formal production QA certification.

15. Technology Stack

Technology	Purpose
Microsoft Power BI	Dashboarding, modelling and visualization
Power Query	Data transformation and preparation
DAX	KPI and analytical calculations

Technology	Purpose
Microsoft Excel	Assessment datasets and structured source files

Other technologies discussed during project planning were not treated as implemented components unless confirmed in the final project.

16. Project Workflow

```
graph TD; A[Assessment Data] --> B[Raw / Staging Data]; B --> C[Power Query Transformation]; C --> D[Cleaning & Standardization]; D --> E[Append / Merge]; E --> F[Dimensions + Fact Tables]; F --> G[Star Schema]; G --> H[DAX Measures]; H --> I[KPI Validation]; I --> J[Dashboard Development]; J --> K[Navigation & Interactions]; K --> L[Visual Refinement]; L --> M[Final Power BI Dashboard];
```

17. Project Outcome

The developed solution provides a centralized Power BI framework for analysing safety, protection and wellbeing programme assessment data.

The final architecture allows users to move from:

State → District → Institution

and separately analyse:

Students → Personnel

The solution combines participation, assessment and learning-outcome information in a single interactive reporting environment.

The dashboard is therefore positioned as a **monitoring and decision-support solution**, rather than simply a collection of charts.

18. Recommendations

Based on the implemented solution, the following recommendations are relevant:

1. Maintain standardized assessment structures across modules.
 2. Maintain consistent participant and institution identifiers.
 3. Preserve raw assessment data separately from transformed analytical data.
 4. Validate Question_ID uniqueness before model relationships.
 5. Maintain actual assessment dates for production reporting.
 6. Use the selected-institution approach consistently across institution pages.
 7. Keep executive reporting separate from detailed student/personnel analysis.
 8. Avoid adding operational KPIs unless corresponding source data exists.
-

19. Future Scope

The following can be considered future enhancements rather than current implemented features:

- Automated data refresh
- Production-scale data integration
- Additional programme modules
- Expanded district/institution coverage
- Automated reporting distribution
- Advanced predictive analytics
- Additional operational monitoring datasets
- More granular intervention tracking
- Production deployment and governance

These should be treated as **future scope**, not current project capabilities.

20. Conclusion

The **Statewide Safety, Protection & Wellbeing Programme Analytics Dashboard** establishes a structured BI solution for analysing programme participation and assessment outcomes across students and personnel.

The project combines structured assessment datasets, Power Query transformation, star-schema modelling and DAX-based analytics to produce interactive state, district and institution-level reporting.

The six-page dashboard architecture separates executive monitoring from detailed student, personnel and institution analysis. The use of geographic mapping, Pre/Post comparison, learning gain, participation analysis, Key Influencers and Decomposition Tree provides multiple levels of analytical exploration.

The resulting solution provides a foundation for centralized programme monitoring and data-driven assessment analysis while maintaining a clear distinction between the current demo implementation and future production-scale enhancements.

PDF DESIGN SPECIFICATION

Page

A4 Portrait

Use landscape pages selectively for:

- Data model
- Dashboard architecture
- Wide dashboard screenshots

Theme

Use the same visual identity as the Power BI dashboard:

- Deep navy/blue for headings
- Light blue background accents
- White content cards
- Subtle grey borders
- Blue/purple accent colors
- Minimal decorative elements

Typography

Recommended:

- **Title:** Aptos Display / Segoe UI

- **Headings:** Aptos / Segoe UI Semibold
- **Body:** Aptos / Segoe UI
- **Tables:** Aptos

Heading hierarchy

```
1. Major Section
1.1 Subsection
1.1.1 Detail
```

Header

Project title shortened to:

Statewide Safety, Protection & Wellbeing Programme

Footer

Programme Analytics Dashboard | Project Report

Page X

Figures

Use:

Figure 1. Final Dashboard Architecture

and below:

Source: Project implementation

Tables

Use:

- Dark blue header
- White text
- Alternating light rows
- Minimal borders

Screenshots

Use only the final/relevant dashboard screenshots.

Recommended figure sequence:

1. Home Dashboard
2. State Executive

3. State Student
 4. State Personnel
 5. Institution Student
 6. Institution Personnel
 7. Final Star Schema
 8. Overall Analytics Workflow
-

PDF GENERATION INSTRUCTIONS

The final PDF should be structured approximately as:

```
Cover
↓
Table of Contents
↓
Executive Summary
↓
Project Overview
↓
Requirements
↓
Datasets
↓
Data Preparation
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Data Model
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DAX & Analytical Measures
↓
KPI Framework
↓
Dashboard Architecture
↓
Page-by-Page Documentation
↓
Visualization & UX
↓
Interactive Features
↓
Analytical Insights
↓
Challenges & Solutions
↓
Validation
↓
Technology Stack
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Project Workflow
```

↓
Project Outcome
↓
Recommendations
↓
Future Scope
↓
Conclusion
↓
Appendices

The uploaded master instructions explicitly require the report to reconstruct the **final project rather than reproduce the conversation**, remove repeated/irrelevant material, distinguish final implementation from proposed ideas, and avoid inventing unavailable information. Pasted text

For the final PDF, the report should therefore use **only confirmed project information**, with anything genuinely unresolved marked as:

[INFORMATION NOT AVAILABLE IN CONVERSATION]

rather than filling the gap with assumptions. Pasted text

Sources